

VIVAAN SINHA

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EDUCATION

California Polytechnic State University, San Luis Obispo
Bachelor of Science (BS) in Computer Science

August 2026 – May 2030
San Luis Obispo, CA

SKILLS

Languages & Tools: Python, C#, Java, HTML5, Pandas, Matplotlib, Jupyter Notebook

ML & Security: Federated Learning, Differential Privacy, Adversarial Defense, Explainable AI (SHAP), Neural Networks

HONORS AND AWARDS

National Semifinalist (Top 40), National Speech and Debate Association, Senate Congressional Debate

RESEARCH EXPERIENCE

University of Texas at Austin

January 2025 - June 2026

Independent Research, advised by Dr. Angela Newell

Austin, Texas

Systems Architecture Paper: Privacy, Robustness, and Explainability in Cybersecurity AI

- Designed a layered cybersecurity-AI architecture integrating federated learning, differential privacy, adversarial defenses, and explainable AI across distinct stages of the ML pipeline; authored a systems-architecture paper outlining the approach.
- Synthesized evidence from 46 peer-reviewed papers, 18 industry reports, and interviews with AWS security professionals to compare technical approaches and design tradeoffs across privacy, robustness, and interpretability.
- Diagnosed a privacy-security tradeoff in federated learning, where privacy mechanisms obscure model behavior and hinder detection of adversarial poisoning attacks; designed a multi-stage explainability and governance layer (TreeSHAP over DP-protected features) to audit training, aggregation, and deployment, restoring visibility without weakening confidentiality.
- Evaluated tradeoffs across five techniques – federated learning, differential privacy, homomorphic encryption, adversarial defenses, and explainable AI – against deployment constraints in critical infrastructure systems.

UC Berkeley Artificial Intelligence Research Lab - Summer Research Lab

June 2025 - July 2025

Predicting Social Media Addiction with Machine and Deep Learning

Berkeley, CA

- Cleaned and engineered features from social media usage data using Python (Pandas) in Jupyter Notebooks and Google Colab, then trained Logistic Regression, Random Forest, and Multi-Layer Perceptron models to predict social media addiction risk.
- Improved Multi-Layer Perceptron accuracy from 96% to 98.6% after refining neural network architecture and feature selection.
- Evaluated classifiers using precision–recall curves, F1 score, and confusion matrices, visualized with Matplotlib, confirming the model reliably identified at-risk users rather than simply predicting "not at-risk" most of the time.
- Analyzed model behavior for bias and failure cases, finding that adding sleep-related features cut false negatives by 18%, improving detection of users actually at risk.

EXPERIENCE

theCoderSchool

August 2025 – August 2026

Programming Instructor

San Ramon, CA

- Designed and delivered project-based instruction in Python, C#, and Unity to 15+ K–8 students through individualized 1-on-1 lessons, tailoring pacing and project complexity to each student's skill level.
- Guided students in building original games, applications, and websites, developing their programming, debugging, and problem-solving skills along the way.
- Selected to lead coding workshops at Vista Grande Elementary and Rancho Romero Elementary schools, expanding access to hands-on computer science education.

United Studios of Self Defense

January 2024 – July 2026

Self Defense Instructor

Danville, CA

- Instructed 50+ students ages 5–19 in a comprehensive martial arts curriculum integrating kata, weapons defense, and boxing.
- Led group classes and 10+ individual training sessions weekly, adapting instruction to students' age, skill level, and development.
- Designed specialized training exercises to strengthen students' technique, discipline, and practical self-defense skills.

Monte Vista High School UNICEF Club

August 2025 – June 2026

President

Danville, CA

- Raised \$950 through student-led fundraisers and partnerships with two local businesses to support humanitarian initiatives.
- Organized two school-wide book drives in partnership with the African Library Project, collecting and shipping 2,500 books to stock libraries in Malawi.
- Led a 60+ member club, planning and organizing meetings 1–2 times monthly and holding weekly check-ins with board members to track project progress.